Advanced Mechanics & Properties of Matter Lab

Lab Record & Viva-Voce Booklet

15 Experiments — printable, fill-by-hand observation, calculation, result and viva sheets

Companion to the interactive simulator and online User Guide

Student Name: ____

Roll / Enrolment No.: ____

Course & Section: ____

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How to Use This Booklet

This booklet mirrors the 15 experiments in the online **User Guide** and the interactive simulator. For each experiment: run the simulator panel as instructed in the guide, record your readings directly in the observation table below, plot your graph on the provided 15×15 grid (choose your own scale per square), complete the calculation and result sections, work out the maximum permissible error using the least counts of the instruments you used, and prepare written answers to the viva-voce questions before your practical examination. Full theoretical background, governing formulas and step-by-step procedure for every experiment are available in the online guide — this booklet intentionally contains only the record-keeping sheets, not the theory, to keep it compact for printing.

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1. Bifilar Suspension — Moment of Inertia of a Rectangular Bar

Aim

To determine the moment of inertia of a rectangular bar about a vertical axis through its centre, using the torsional oscillations of a bifilar suspension.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$I = M g b^2 T^2 / (4\pi^2 l)$$

M = mass of the bar
b = half-separation between the two threads
T = period of torsional oscillation
l = length of each thread

Labelled diagram of the experimental apparatus:

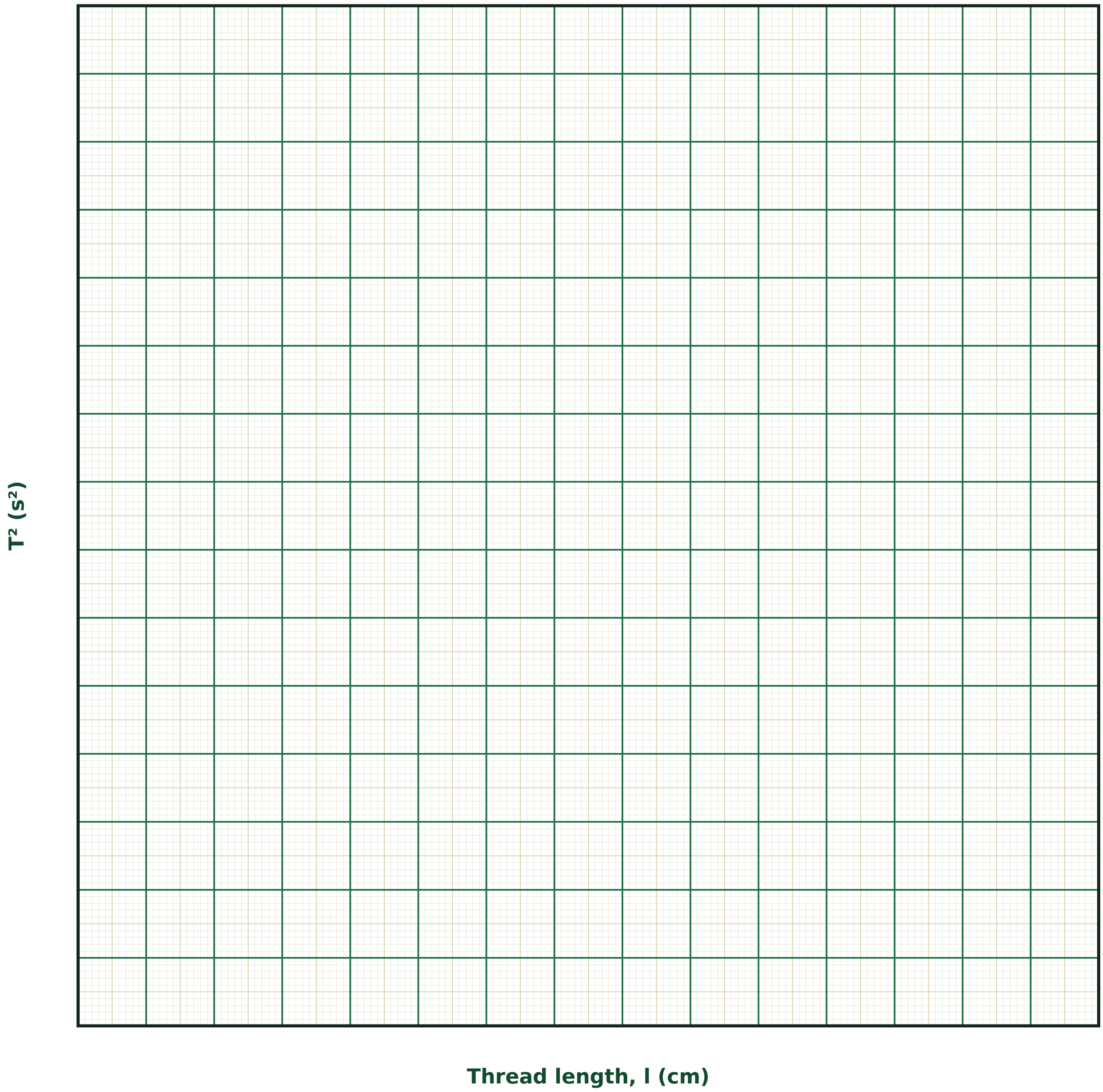
Observation Table

S. No.	l (cm)	b (cm)	M (g)	Period T (s) — observed	T^2 (s^2)

S. No.	l (cm)	b (cm)	M (g)	Period T (s) — observed	T^2 (s^2)

Graph

T^2 vs Thread Length



Calculation

Result

Trial	Slope of T^2 vs l (s^2/cm)	M (g)	b (cm)	I ($g \cdot cm^2$)

Maximum Permissible Error

Formula: $\Delta I / I = \Delta M/M + 2 \cdot \Delta b/b + 2 \cdot \Delta T/T + \Delta l/l$

$I = Mgb^2T^2/(4\pi^2 l)$ is a product/quotient of measured quantities, with b and T entering squared — their fractional errors are doubled before summing.

Quantity	Measured Value	Least Count (LC)
M — Mass of bar (g)		
b — Half-separation (cm)		
T — Period (s)		
l — Thread length (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why must the two threads be exactly parallel and equal in length before starting the experiment?

2. How would the period change if the bar's mass were doubled but its length (hence moment of inertia per unit mass) stayed the same?

3. Derive, in outline, why the restoring torque is proportional to the angular displacement for small twists.

4. What practical advantage does the bifilar method have over a torsional pendulum using a single twisted wire?

5. How would you extend this method to find the moment of inertia of an irregularly shaped object?

6. Why is it important to time many oscillations rather than a single one?

2. Torsional Pendulum — Rigidity Modulus and Moment of Inertia by the Two-Body Method

Aim

To determine the rigidity (shear) modulus of the material of a suspension wire and the moment of inertia of a circular disc, using the two-body method of a torsional pendulum.

Date performed: ___________________   Simulator panel used: _____________________________

Formula Used

$$I = 2md^2T^2 / (T'^2 - T^2) \; ; \; \eta = 8\pi I l / (r^4 T^2)$$

m, d = each added mass and its distance from the axis
T, T′ = period of the disc alone, and with the two added masses
l, r = length and radius of the suspension wire

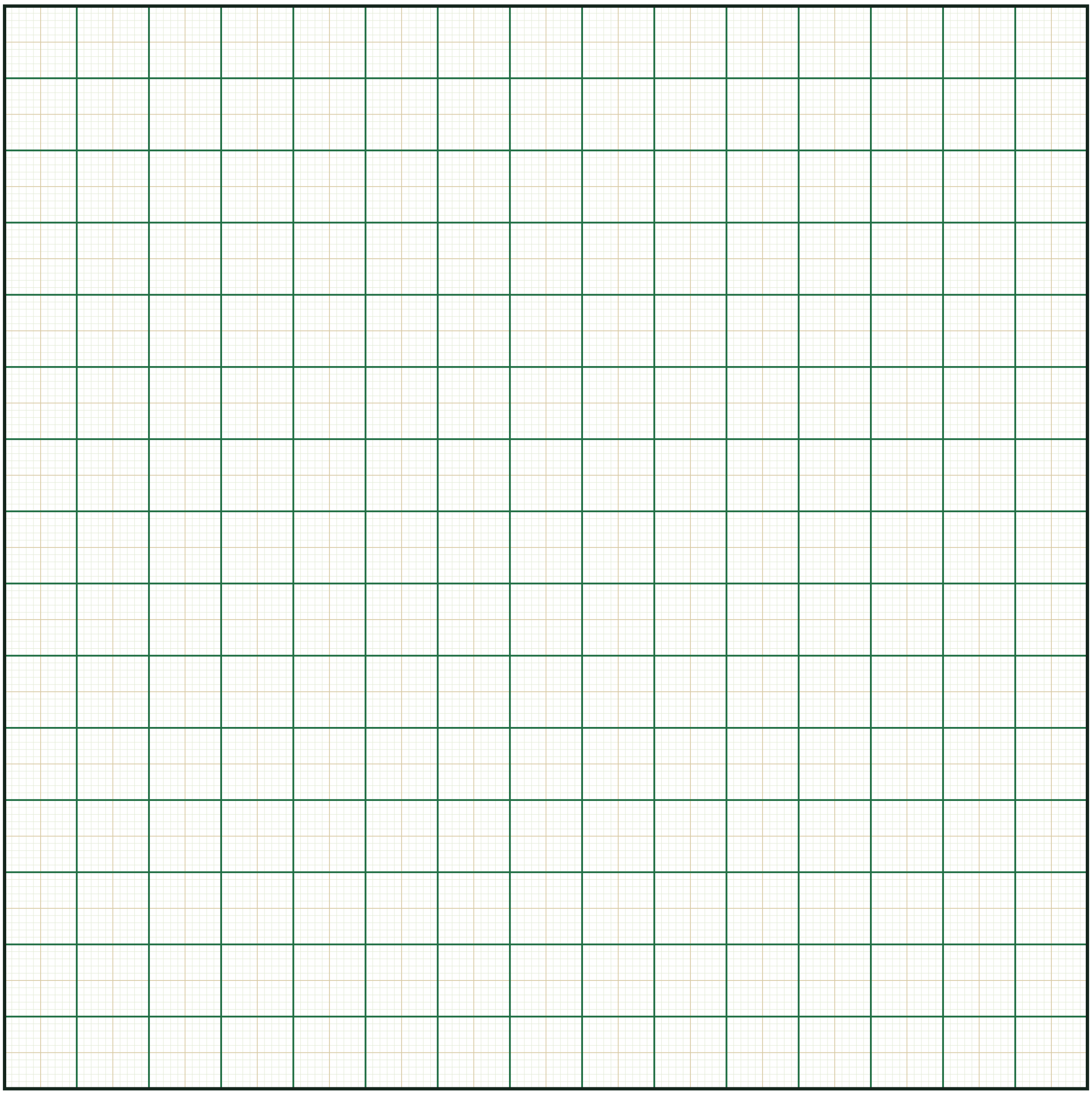
Labelled diagram of the experimental apparatus:

Observation Table

S. No.	d (cm)	m (g)	T (s)	T′ (s)	$I = 2md^2T^2/(T'^2-T^2)$ (g·cm²)

S. No.	d (cm)	m (g)	T (s)	T′ (s)	$I = 2md^2T^2/(T'^2 - T^2)$ ($g \cdot cm^2$)

Graph



Calculation

Result

Trial	l (cm)	r (mm)	I (g·cm²)	η (dyne/cm²)

Maximum Permissible Error

Formula: $\Delta\eta / \eta = \Delta l/l + \Delta I/I + 4 \cdot \Delta r/r + 2 \cdot \Delta T/T$

$\eta = 8\pi I l/(r^4 T^2)$ depends on r to the fourth power and T squared — even a small uncertainty in the wire radius (measured with a screw gauge) dominates the total error.

Quantity	Measured Value	Least Count (LC)
l — Wire length (cm)		
I — Moment of inertia of disc (g·cm²)		
r — Wire radius (mm)		
T — Period (disc alone) (s)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does the rigidity modulus depend on the fourth power of the wire radius rather than, say, the square?

2. What is the two-body method designed to eliminate, and how does it achieve this algebraically?

3. Distinguish between Young's modulus and the rigidity (shear) modulus — what kind of deformation does each describe?

4. Why must the two added masses be placed at exactly equal distances from the axis and be of exactly equal mass?

5. How would you check experimentally that the wire is genuinely obeying Hooke's law throughout the twist range you used?

6. What would happen to the period if the disc were replaced by a ring of the same mass and radius?

3. Flywheel — Moment of Inertia and Frictional Couple

Aim

To determine the moment of inertia of a flywheel about its own axis, and the frictional couple opposing its rotation, by the falling-mass method.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$I = m\,r^2(2gh - \omega^2 r^2)/\omega^2 \times n_2/(n_1+n_2)$$

m = falling mass
r = radius of the axle
h = height of fall
ω = angular velocity when the cord leaves the axle
n_1, n_2 = cord turns on the axle, and extra revolutions before stopping

Labelled diagram of the experimental apparatus:

Observation Table

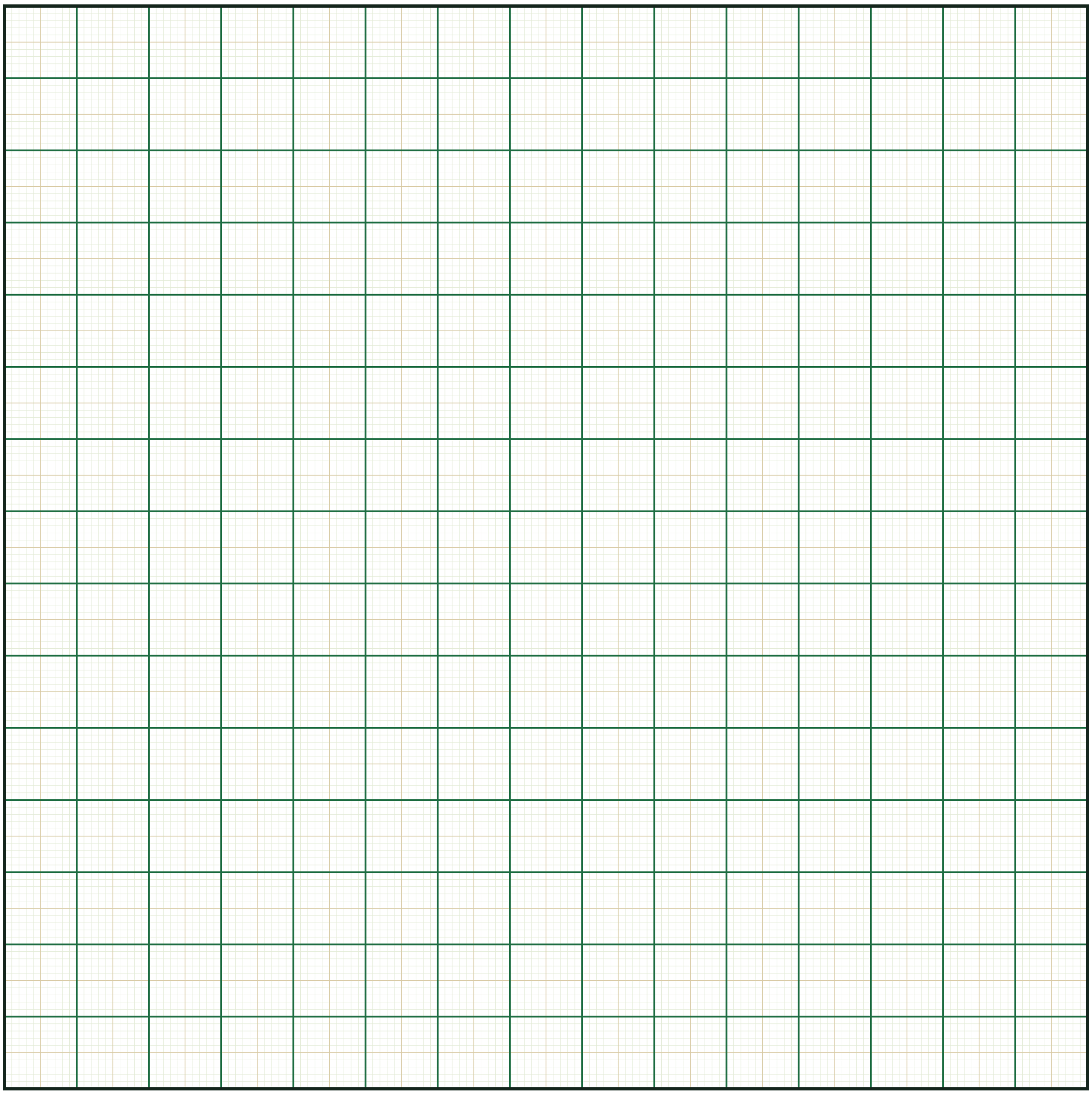
S. No.	m (g)	h (cm)	t (s)	n_2 (rev)	$\omega = 2\pi n_1/t$ (rad/s)	I (g·cm²)

S. No.	m (g)	h (cm)	t (s)	n_2 (rev)	$\omega = 2\pi n_1/t$ (rad/s)	I ($g \cdot cm^2$)

Graph

$1/t^2$ vs Falling Mass

$1/t^2$ (s^{-2})



Falling mass, m (g)

Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta I/I \approx 2\cdot\Delta r/r + \Delta h/h + 2\cdot\Delta t/t + \Delta n_2/n_2$

The axle radius r and fall time t both enter squared in the governing formula, so their fractional errors are doubled before summing with the others.

Quantity	Measured Value	Least Count (LC)
r — Axle radius (cm)		
h — Height of fall (cm)		
t — Fall time (s)		
n_2 — Extra revolutions (rev)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is the falling-mass method able to determine both I and the frictional torque without an independent friction measurement?

2. What assumption is made about the frictional torque during the two phases of the motion, and how reasonable is it?

3. How would you experimentally locate the moment when the cord just leaves the axle?

4. Why is it important to measure n_2 (extra revolutions) as well as the fall time t?

5. How would the result change if the cord were wound in more than one layer?

6. Why do larger flywheels (larger I) take longer to both spin up and spin down?

4. Compound (Bar) Pendulum — Acceleration due to Gravity and Radius of Gyration

Aim

To determine the acceleration due to gravity g and the radius of gyration k of a uniform bar pendulum, using the method of conjugate points.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$T = 2\pi\sqrt{[(k^2+h^2)/gh]} \; ; \; g = 4\pi^2 L/T^2$$

h = distance of the pivot from the centre of gravity
k = radius of gyration about the centre of gravity
L = equivalent simple-pendulum length (sum of conjugate distances)

Labelled diagram of the experimental apparatus:

Observation Table

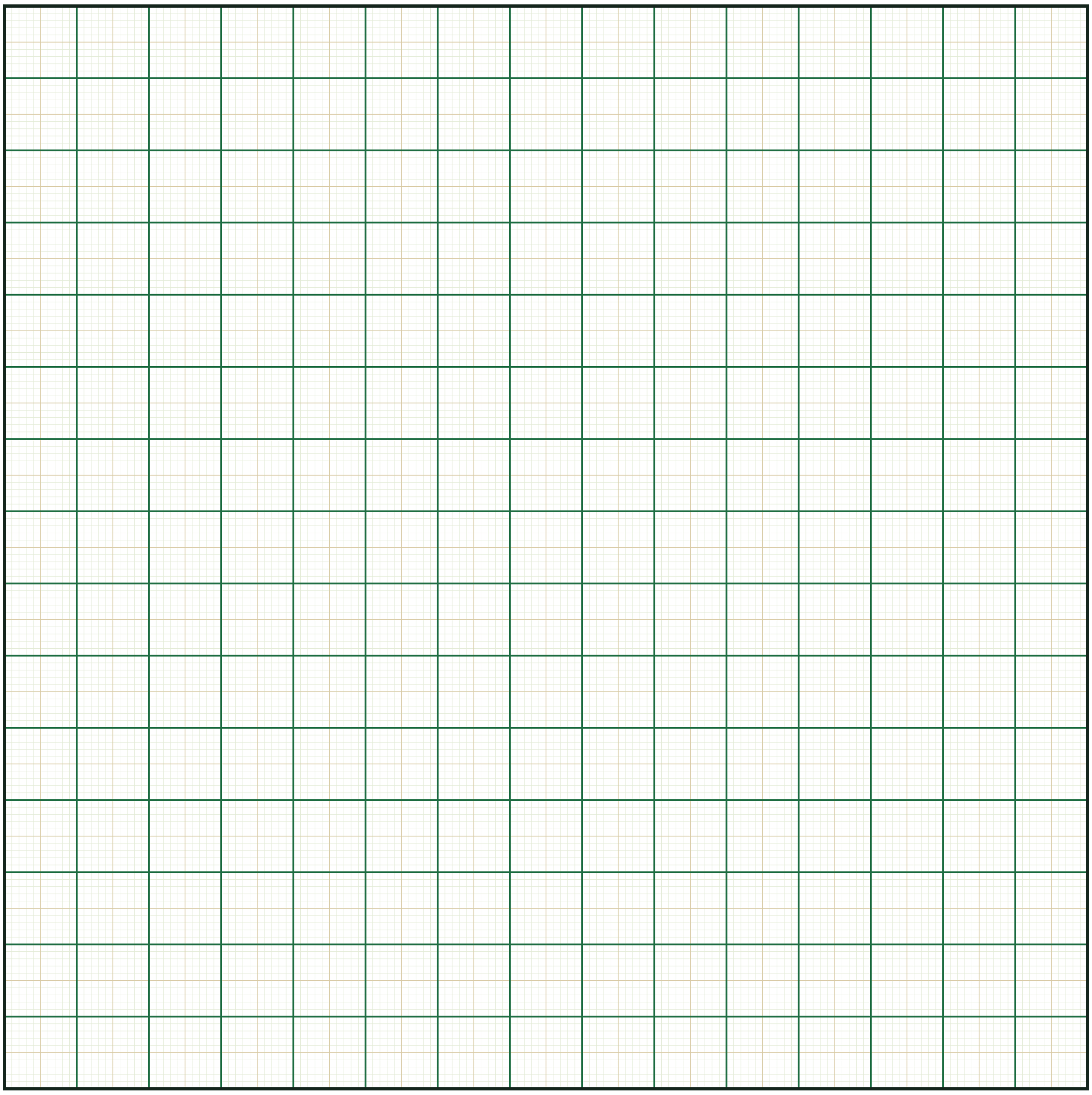
S. No.	h (cm)	Period T (s)	T^2 (s^2)

S. No.	h (cm)	Period T (s)	T^2 (s^2)

Graph

Period vs Pivot Distance

Period, T (s)



Pivot distance, h (cm)

Calculation

Result

Conjugate pair	h_1 (cm)	h_2 (cm)	T (s)	g (cm/s²)	k (cm)

Maximum Permissible Error

Formula: $\Delta g/g = \Delta L/L + 2\cdot\Delta T/T$

$g = 4\pi^2 L/T^2$ depends on T squared, so timing precision (usually via timing many oscillations) dominates the error budget more than the length measurement L.

Quantity	Measured Value	Least Count (LC)
L — Equivalent length (h■+h■) (cm)		
T — Period (s)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does the compound pendulum have a minimum period at some pivot distance rather than at the centre of gravity itself?

2. What physically defines a pair of conjugate points, and why must $L = h_1+h_2$ equal the equivalent simple pendulum length?

3. How is Kater's reversible pendulum related to this experiment, and why was it historically important?

4. Why would timing 20 or 50 oscillations, rather than one, substantially improve the accuracy of g?

5. How would the T vs h curve change if the bar were replaced by one of the same length but different mass distribution (e.g. with added end masses)?

6. What would happen to the period if the bar were pivoted exactly at its centre of gravity?

5. Non-Uniform (Cantilever) Bending — Young's Modulus

Aim

To determine the Young's modulus of the material of a beam by the method of non-uniform (cantilever) bending.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$Y = 4Wl^3 / (bd^3y)$

W = load applied at the free end
l = length of the cantilever (clamp to load point)
b, d = breadth and thickness of the beam
y = depression of the free end

Labelled diagram of the experimental apparatus:

Observation Table

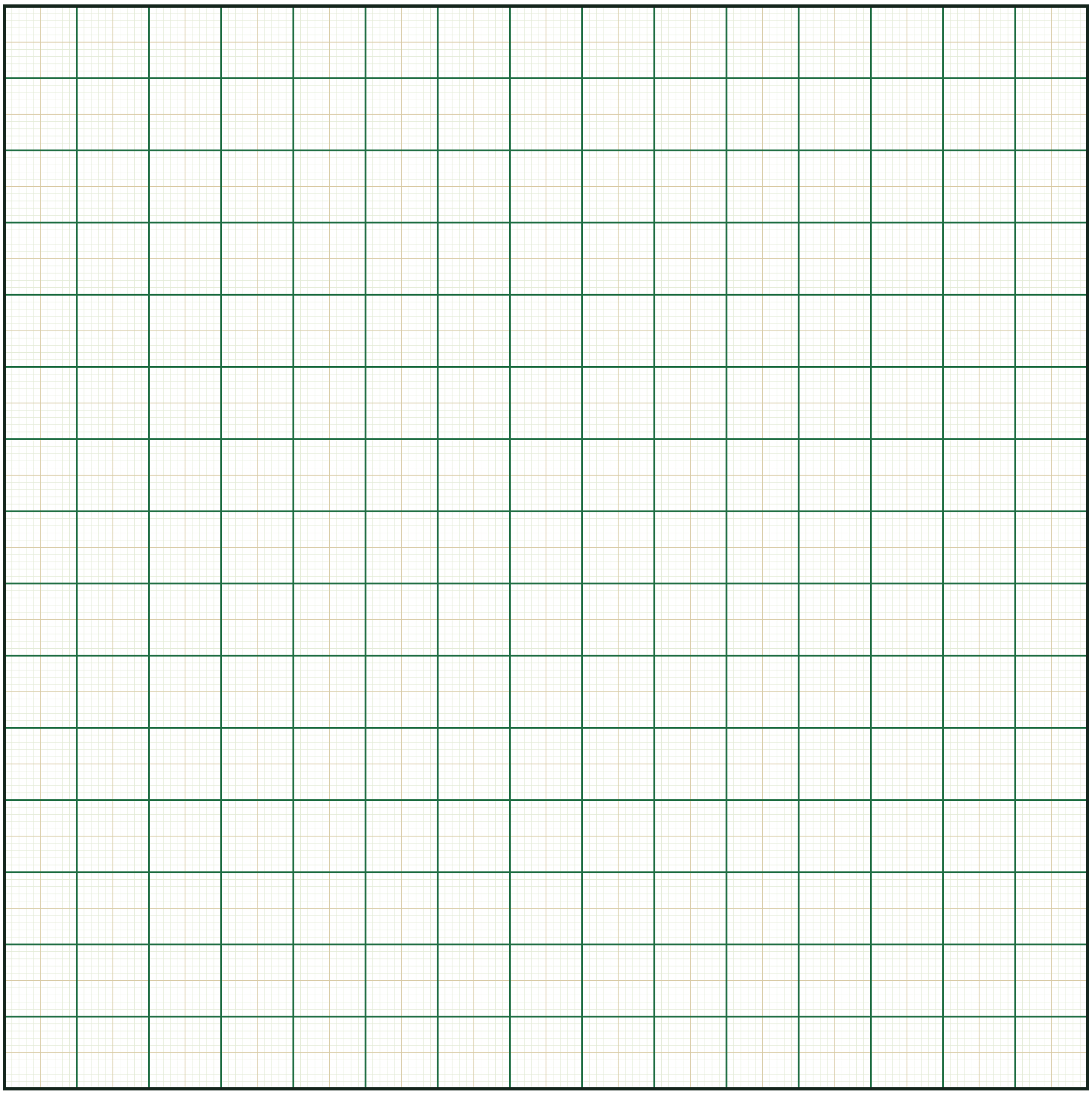
S. No.	M (g)	l (cm)	b (cm)	d (cm)	Depression y (mm)

S. No.	M (g)	l (cm)	b (cm)	d (cm)	Depression y (mm)

Graph

Depression vs Load

Depression, y (mm)



Load, M (g)

Calculation

Result

Trial	l (cm)	b (cm)	d (cm)	Slope (mm/g)	Y (dyne/cm²)

Maximum Permissible Error

Formula: $\Delta Y/Y = \Delta M/M + 3\cdot\Delta l/l + \Delta b/b + 3\cdot\Delta d/d + \Delta y/y$

$Y = 4Wl^3/(bd^3y)$ depends on the cantilever length l cubed and on the thickness d cubed (inversely) — small errors in measuring l or, especially, the thin dimension d, are amplified threefold.

Quantity	Measured Value	Least Count (LC)
M — Load (g)		
l — Cantilever length (cm)		
b — Breadth (cm)		
d — Thickness (cm)		
y — Depression (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is this bending called 'non-uniform', and how does that differ from the uniform bending experiment?

2. Where along the neutral axis does a longitudinal filament experience zero strain, and why?

3. Why does the depression depend on the cube of the length but only the cube of the thickness (inversely), rather than some other power?

4. What experimental precautions minimise systematic error from the beam's own weight and any residual bending before loading?

5. How would you distinguish, experimentally, whether a beam is behaving elastically (returns to zero on unloading) or has undergone some permanent set?

6. Why is a travelling microscope, rather than a ruler, used to measure the depression?

6. Uniform Bending — Young's Modulus by the Elevation Method

Aim

To determine the Young's modulus of the material of a beam by the method of uniform bending, using symmetric end-loading and measuring the elevation of the midpoint.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$Y = 3Mga\,l^2 / (2bd^3e)$$

M = each of the two equal end loads
a = overhang from each knife edge to its load point
l = separation between the two knife edges
e = elevation of the midpoint

Labelled diagram of the experimental apparatus:

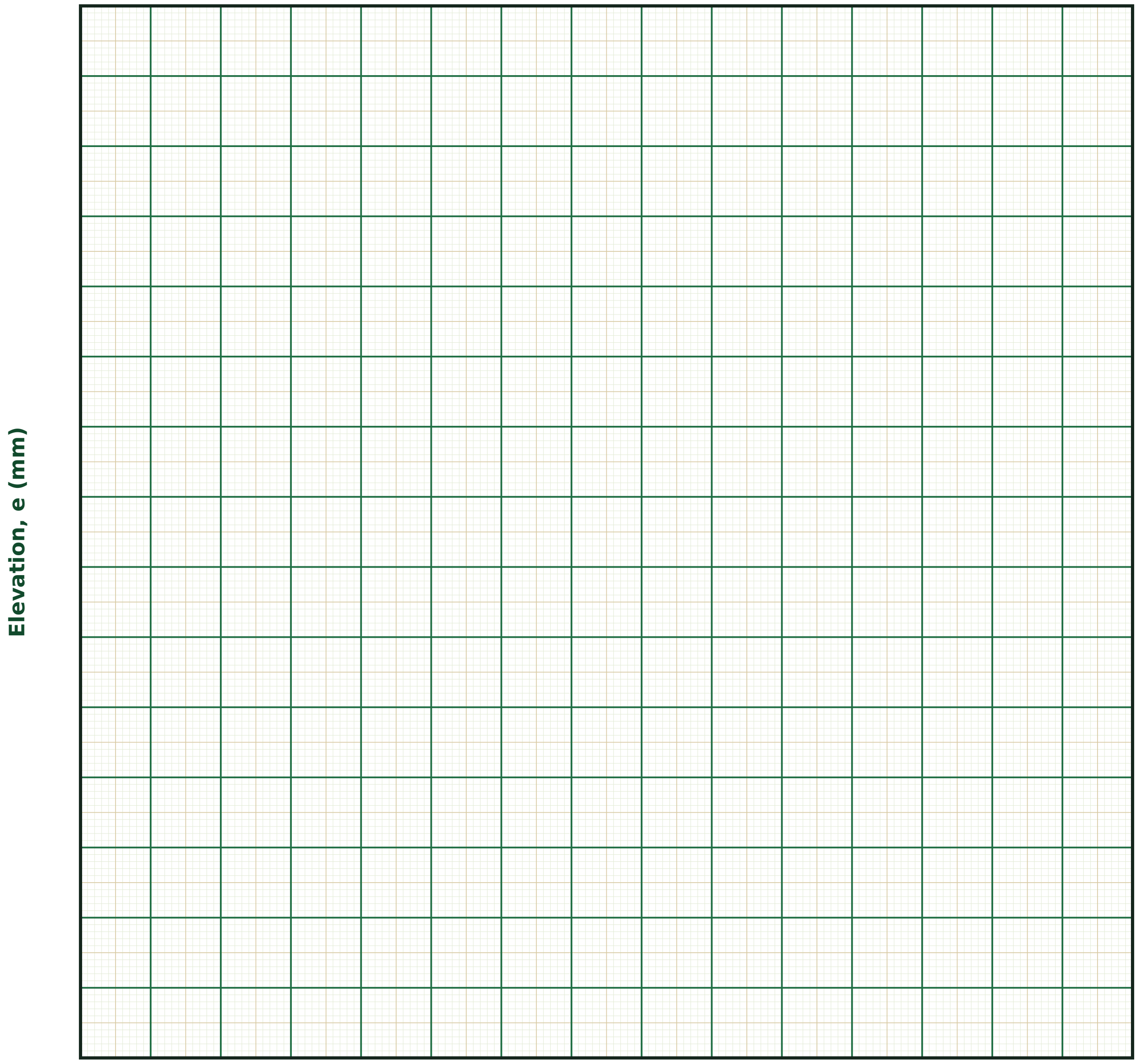
Observation Table

S. No.	M (g)	a (cm)	l (cm)	Elevation e (mm)

S. No.	M (g)	a (cm)	l (cm)	Elevation e (mm)

Graph

Elevation vs End Load



Calculation

Result

Trial	a (cm)	l (cm)	Slope (mm/g)	Y (dyne/cm²)

Maximum Permissible Error

Formula: $\Delta Y/Y = \Delta M/M + \Delta a/a + 2\cdot\Delta l/l + \Delta b/b + 3\cdot\Delta d/d + \Delta e/e$

As in the cantilever method, the beam thickness d enters cubed, so its measurement uncertainty (via a screw gauge) usually dominates the total error.

Quantity	Measured Value	Least Count (LC)
M — Each end load (g)		
a — Overhang (cm)		
l — Knife-edge separation (cm)		
b — Breadth (cm)		
d — Thickness (cm)		
e — Elevation (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is the bending moment constant between the knife edges in this experiment, whereas it varies along a cantilever?

2. What experimental conditions are essential to ensure the 'uniform' bending assumption actually holds?

3. Why does the elevation depend on the square of the knife-edge separation, rather than its cube as in the cantilever case?

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4. How would you check, using your own data, whether the beam is within its elastic limit throughout the range of loads used?

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5. Compare the sensitivity of Y to the thickness d in this method versus the cantilever method.

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6. Why is symmetric loading (equal M, equal a on both sides) essential to the validity of the formula used?

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7. Searle's Apparatus — Young's Modulus by Direct Elongation

Aim

To determine the Young's modulus of the material of a wire by directly measuring its elongation under a known tensile load, using Searle's apparatus.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$Y = MgL / (\pi r^2 l_{ext})$

L = original (unstretched) length of the wire
r = radius of the wire
l_ext = elongation under load Mg

Labelled diagram of the experimental apparatus:

Observation Table

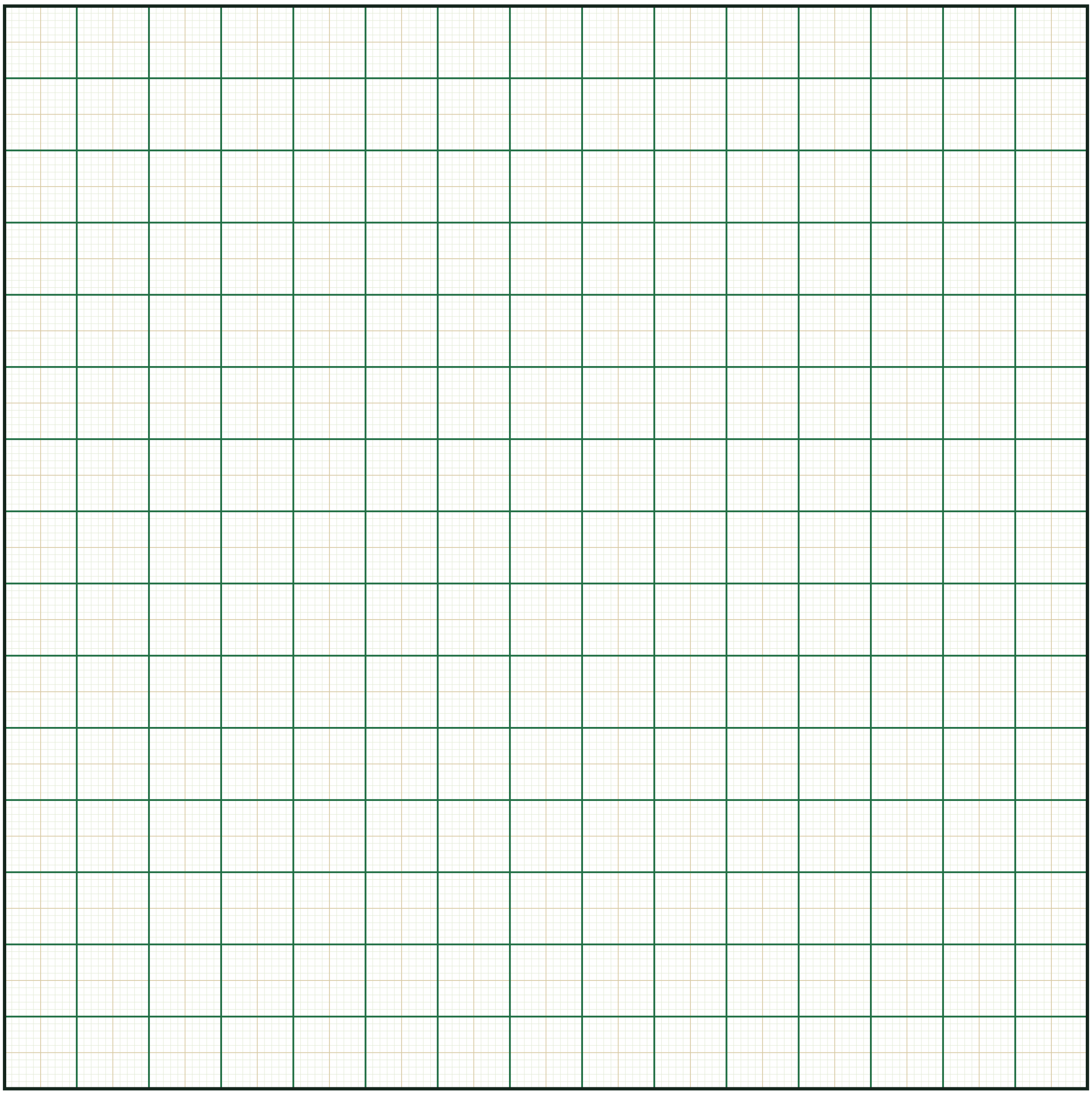
S. No.	Load M (g)	Micrometer reading — loading (mm)	Micrometer reading — unloading (mm)	Mean elongation (mm)

S. No.	Load M (g)	Micrometer reading — loading (mm)	Micrometer reading — unloading (mm)	Mean elongation (mm)

Graph

Elongation vs Load

Elongation (mm)



Load, M (g)

Calculation

Result

L (cm)	r (mm)	Slope (mm/g)	Y (dyne/cm²)

Maximum Permissible Error

Formula: $\Delta Y/Y = \Delta M/M + \Delta L/L + 2\cdot\Delta r/r + \Delta l_{ext}/l_{ext}$

The wire radius r, measured with a screw gauge, enters squared and is typically sub-millimetre — its fractional error usually dominates the overall uncertainty in Y.

Quantity	Measured Value	Least Count (LC)
M — Load (g)		
L — Wire length (cm)		
r — Wire radius (mm)		
l_ext — Elongation (mm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does Searle's apparatus use two wires rather than one, and what specific systematic errors does this design eliminate?

2. Derive the relation between stress, strain and Young's modulus, and identify each in this specific experiment.

3. Why is it important to remove kinks and apply an initial small load before beginning the actual measurement?

4. What is backlash error in a micrometer screw, and how is it avoided in this experiment?

5. How would you verify, purely from your own data, that the wire remained within its elastic limit throughout?

6. Why is the wire radius, rather than its length, usually the dominant source of error in the final result for Y?

8. Static Torsion — Rigidity Modulus of a Wire by the Static Twist Method

Aim

To determine the rigidity (shear) modulus of the material of a wire by measuring its static angle of twist under a known applied couple.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\theta = 2Cl / (\pi\eta r^4)$$

C = applied couple, $C = F \times R_{arm}$
l, r = length and radius of the wire
θ = resulting angle of twist (radians)

Labelled diagram of the experimental apparatus:

Observation Table

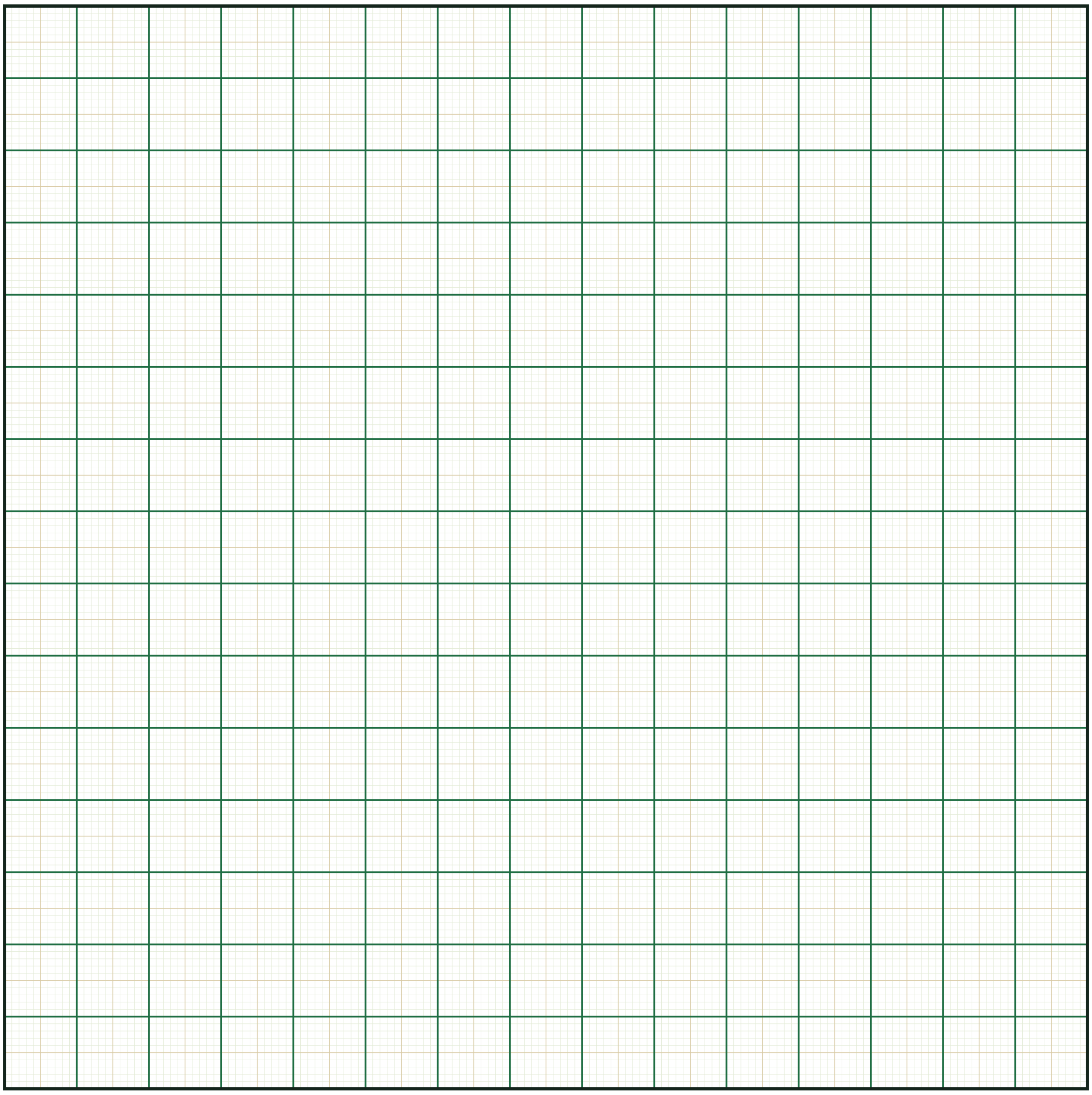
S. No.	F (g)	R_arm (cm)	C = F·R_arm (g·cm)	θ (°)	θ (rad)

S. No.	F (g)	R_arm (cm)	C = F·R_arm (g·cm)	θ (°)	θ (rad)

Graph

Twist Angle vs Applied Couple

Twist angle, θ (rad)



Applied couple, C (g·cm)

Calculation

Result

l (cm)	r (mm)	Slope (rad per g·cm)	η (dyne/cm²)

Maximum Permissible Error

Formula: $\Delta\eta/\eta = \Delta C/C + \Delta l/l + 4\cdot\Delta r/r + \Delta\theta/\theta$

As in the torsional pendulum, the wire radius r enters raised to the fourth power, so its measurement error (from a screw gauge) dominates the total uncertainty in η by far.

Quantity	Measured Value	Least Count (LC)
C — Applied couple (g·cm)		
l — Wire length (cm)		
r — Wire radius (mm)		
θ — Twist angle (rad)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive, in outline, why the twist angle is directly proportional to the applied couple for an elastic wire.

2. Why does the twist angle depend on the fourth power of the wire radius?

3. What experimental precautions ensure that the applied couple is accurately F·R_arm and nothing else?

4. How would you check, using only your own loading/unloading data, whether the wire remained within its elastic limit?

5. Compare the static torsion method with the torsional pendulum method for determining η — what are the relative advantages of each?

6. Why is it important for the cord to leave the pulley exactly horizontally?

9. Poiseuille's Method — Coefficient of Viscosity of a Liquid

Aim

To determine the coefficient of viscosity of a liquid by studying its steady, streamline flow through a narrow capillary tube under a constant pressure head.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\eta = \pi P r^4 t / (8Vl) \; ; \; P = \rho g h$$

P = driving pressure (liquid column head h)
r, l = radius and length of the capillary
V, t = volume collected in time t

Labelled diagram of the experimental apparatus:

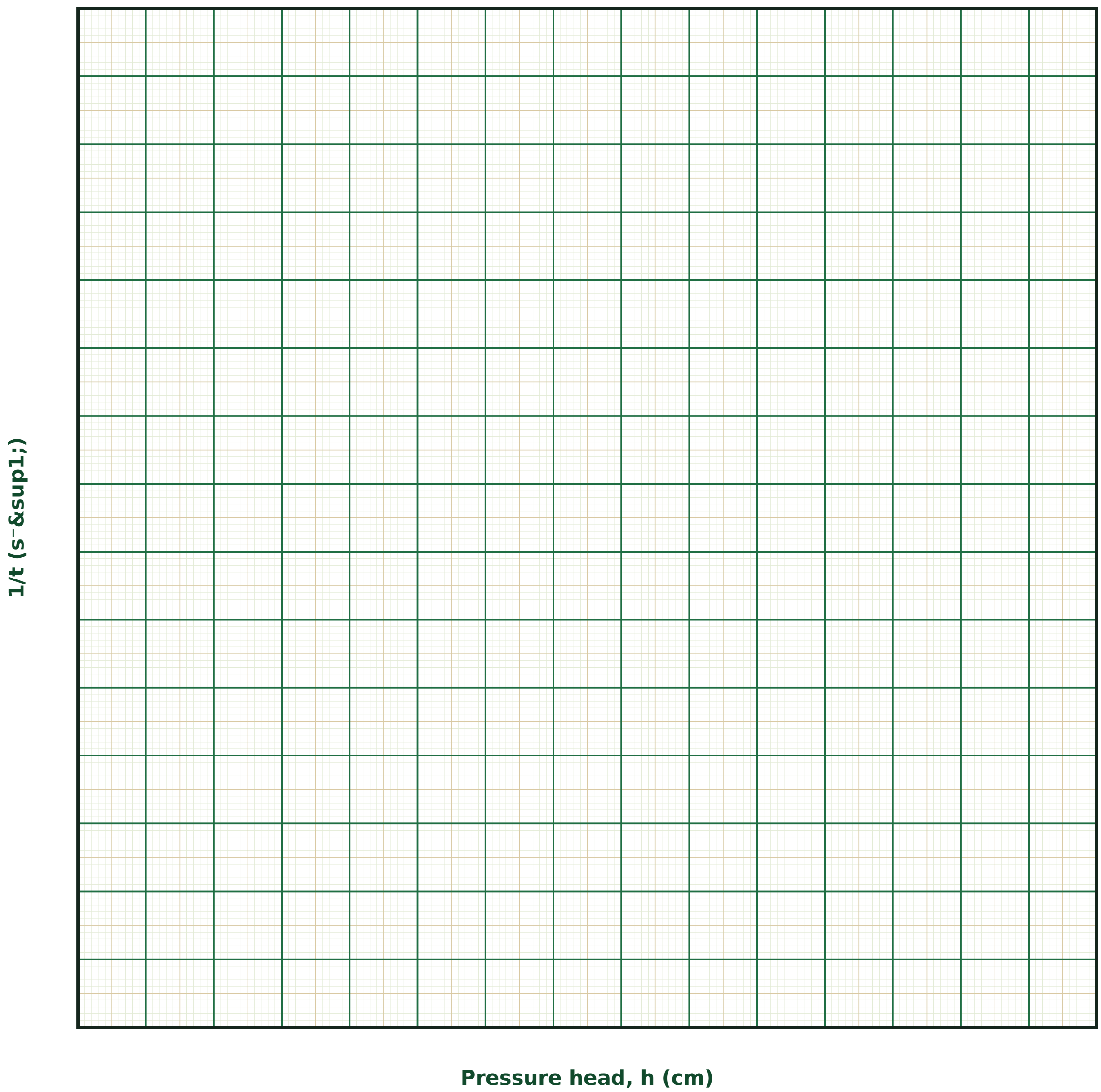
Observation Table

S. No.	h (cm)	Volume V (cm^3)	Time t (s)	1/t (s^{-1})

S. No.	h (cm)	Volume V (cm^3)	Time t (s)	1/t (s^{-1})

Graph

1/t vs Pressure Head



Calculation

Result

Liquid	ρ (kg/m³)	r (mm)	l (cm)	Slope (s⁻¹/cm)	η (poise)

Maximum Permissible Error

Formula: $\Delta\eta/\eta = \Delta h/h + 4\cdot\Delta r/r + \Delta l/l + \Delta V/V + \Delta t/t$

η depends on the capillary radius to the fourth power — even a very small uncertainty in r, typically measured with a travelling microscope on a mercury thread, dominates the total error.

Quantity	Measured Value	Least Count (LC)
h — Pressure head (cm)		
r — Capillary radius (mm)		
l — Capillary length (cm)		
V — Volume collected (cm^3)		
t — Time (s)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is the flow rate through a capillary so extremely sensitive to its radius, compared to its length?

2. What physical assumption distinguishes streamline (laminar) flow from turbulent flow, and how would you recognise experimentally that the flow has become turbulent?

3. Why must the pressure head be kept constant during the timed collection, rather than allowed to fall as the reservoir empties?

4. How would the required collection time change if a more viscous liquid (e.g. glycerine instead of water) were used, all else equal?

5. Derive, in outline, why the velocity profile across the capillary is parabolic in laminar flow.

6. What everyday or industrial situations depend critically on the r^4 law demonstrated in this experiment?

10. Stokes' Method — Viscosity by Terminal Velocity of a Falling Sphere

Aim

To determine the coefficient of viscosity of a highly viscous liquid by measuring the terminal velocity of small spheres falling freely through it.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\eta = (2/9)\, r^2 g(\rho - \sigma)/v$$

r = radius of the falling sphere
ρ, σ = density of the sphere and of the liquid
v = terminal velocity

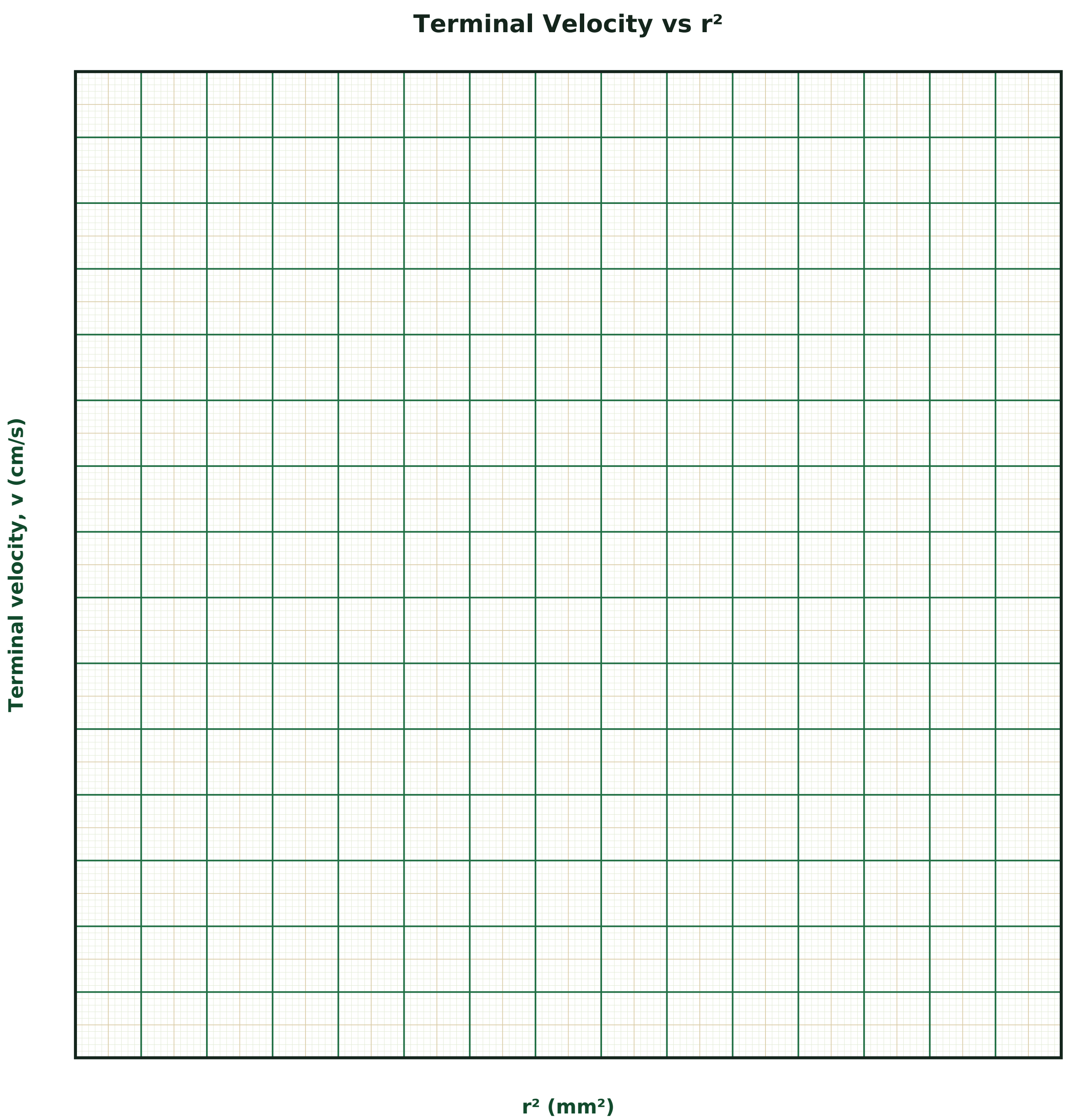
Labelled diagram of the experimental apparatus:

Observation Table

S. No.	r (mm)	r^2 (mm^2)	ρ (kg/m^3)	Terminal velocity v (cm/s)

S. No.	r (mm)	r^2 (mm^2)	ρ (kg/m^3)	Terminal velocity v (cm/s)

Graph



Calculation

Result

Liquid	σ (kg/m³)	Sphere material	ρ (kg/m³)	Slope (cm/s per mm²)	η (poise)

Maximum Permissible Error

Formula: $\Delta\eta/\eta = 2\cdot\Delta r/r + \Delta v/v + (\Delta\rho+\Delta\sigma)/(\rho-\sigma)$

The sphere radius enters squared, and since ρ and σ are usually close in value for typical sphere/liquid combinations, the small difference $(\rho-\sigma)$ can itself carry a relatively large fractional uncertainty.

Quantity	Measured Value	Least Count (LC)
r — Sphere radius (mm)		
v — Terminal velocity (cm/s)		
ρ — Sphere density (kg/m³)		
σ — Liquid density (kg/m³)		
$\Delta(\rho-\sigma)$ — Uncertainty in $(\rho-\sigma)$ (kg/m³)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. State Stokes' law and identify the three forces that act on a falling sphere in a viscous liquid.

2. Why must the sphere be released well above the marked timing points?

3. What correction becomes necessary if the sphere's diameter is not small compared to the diameter of the containing jar?

4. How would the terminal velocity change if a sphere of the same radius but greater density were used?

5. Why does this method work well for glycerine but poorly for water?

6. What is the Reynolds number, and why does Stokes' law require it to be small?

11. Capillary Rise Method — Surface Tension of a Liquid

Aim

To determine the surface tension of a liquid by measuring the height to which it rises in a capillary tube of known bore.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$T = h\,r\,\rho\,g / 2$

h = height of capillary rise
r = internal radius of the capillary
ρ = density of the liquid

Labelled diagram of the experimental apparatus:

Observation Table

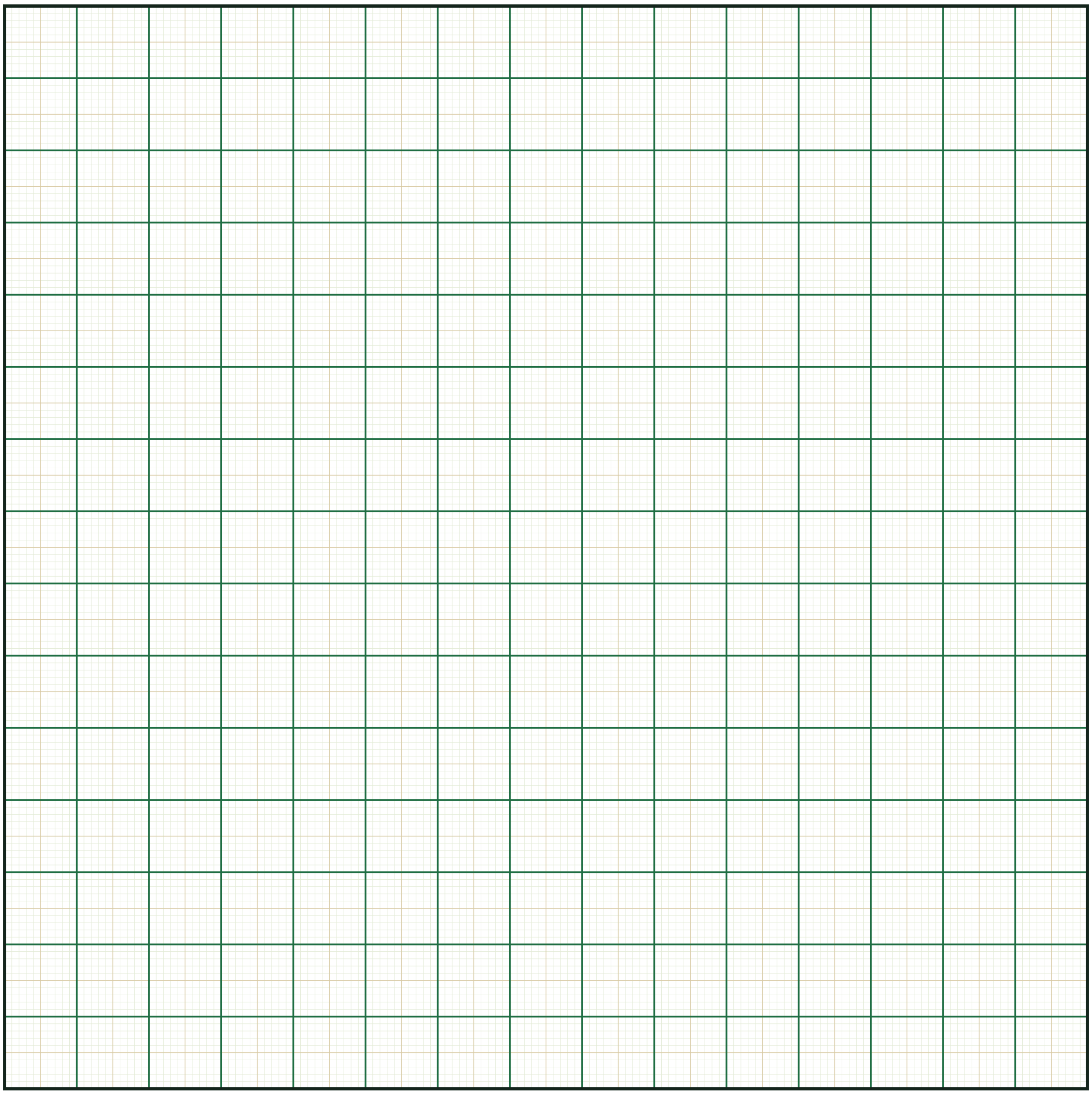
S. No.	r (mm)	1/r (mm^{-1})	ρ (kg/m^3)	Rise height h (cm)

S. No.	r (mm)	1/r (mm^{-1})	ρ (kg/m^3)	Rise height h (cm)

Graph

Rise Height vs 1/r

Rise height, h (cm)



1/r (mm^{-}¹)

Calculation

Result

Liquid	ρ (kg/m³)	Slope (cm·mm)	T (dyne/cm)

Maximum Permissible Error

Formula: $\Delta T/T = \Delta h/h + \Delta r/r$

Both h and r are lengths measured with a travelling microscope, so their combined fractional error is usually small — but the meniscus curvature must be read to the same point (bottom of meniscus) consistently to avoid a systematic bias.

Quantity	Measured Value	Least Count (LC)
h — Rise height (cm)		
r — Capillary radius (mm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does a wetting liquid rise in a capillary tube while a non-wetting liquid (like mercury) is depressed?

2. Derive, in outline, the relation $h = 2T/(r\rho g)$ by balancing forces on the raised liquid column.

3. Why is it advantageous to use several capillaries of different bore and plot h vs 1/r, rather than relying on a single measurement?

4. What effect would a greasy or contaminated capillary have on the observed rise, and why?

5. How does temperature generally affect the surface tension of a liquid?

6. Why must the capillary be held exactly vertical during the measurement?

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12. Coupled Pendulums — Normal Modes and Beats

Aim

To study the normal modes of two identical pendulums coupled by a light spring, and to observe and interpret the phenomenon of beats arising from energy transfer between them.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$f_s=(1/2\pi)\sqrt{(g/l)}\ ;\ f_a=(1/2\pi)\sqrt{(g/l+2k/m)}\ ;\ T_{beat}=1/(f_a-f_s)$$

l = length of each (identical) pendulum
k = coupling spring constant
m = bob mass

Labelled diagram of the experimental apparatus:

Observation Table

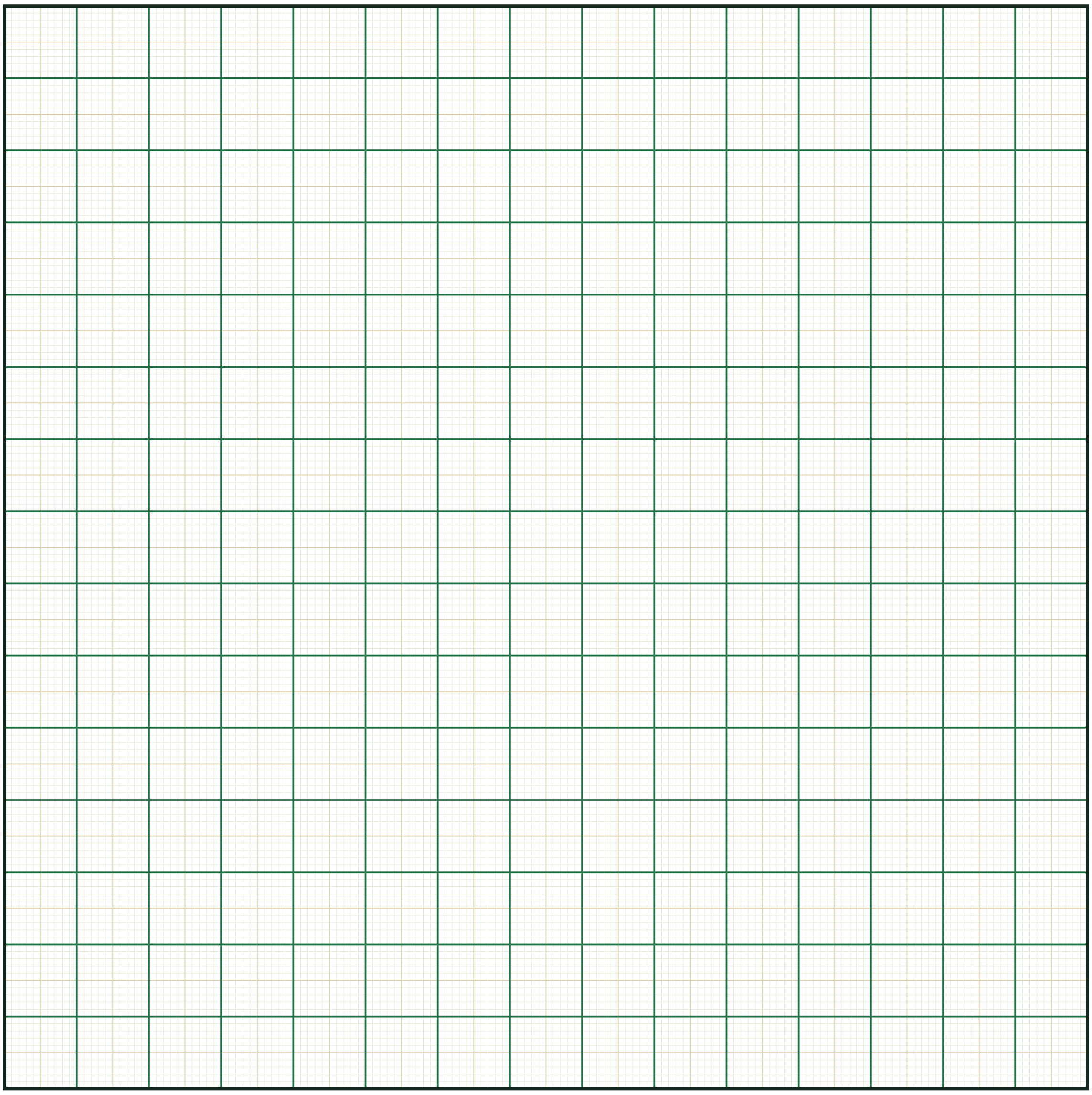
S. No.	l (cm)	k (N/m)	m (g)	f_s (Hz)	f_a (Hz)	T_beat (s)

S. No.	l (cm)	k (N/m)	m (g)	f_s (Hz)	f_a (Hz)	T_beat (s)

Graph

Beat Period vs Coupling Constant

Beat period, T_beat (s)



Coupling spring constant, k (N/m)

Calculation

Result

l (cm)	k (N/m)	f_s (Hz)	f_a (Hz)	T_beat, calculated (s)	T_beat, observed (s)

Maximum Permissible Error

Formula: $\Delta T_{beat}/T_{beat} \approx (\Delta f_a + \Delta f_s)/(f_a - f_s)$

Because T_{beat} depends on the small difference $(f_a - f_s)$, even modest uncertainties in either frequency are amplified when this difference is small (weak coupling).

Quantity	Measured Value	Least Count (LC)
f_a — Antisymmetric mode frequency (Hz)		
f_s — Symmetric mode frequency (Hz)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Explain physically why releasing just one pendulum excites an equal superposition of both normal modes.

2. Why does the symmetric mode frequency not depend on the coupling spring constant at all?

3. Derive, in outline, the beat period in terms of the two normal-mode frequencies.

4. What would happen to the beating pattern if the two pendulums had noticeably different lengths?

5. Name at least one other physical system (electrical, molecular, or optical) that shows an analogous normal-mode beating phenomenon.

6. How would friction/damping, if included, eventually affect the long-term behaviour of this coupled system?

13. Damped Harmonic Oscillations — Logarithmic Decrement and Quality Factor

Aim

To study the decay of amplitude in a damped harmonic oscillator, and to determine the logarithmic decrement and quality factor characterising the damping.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\delta = \ln(A_n/A_{(n+1)}) = \gamma T \; ; \; Q \approx \pi/\delta$$

A_n, $A_{(n+1)}$ = two successive peak amplitudes
γ = damping constant (= b/2m)
T = period of the damped oscillation

Labelled diagram of the experimental apparatus:

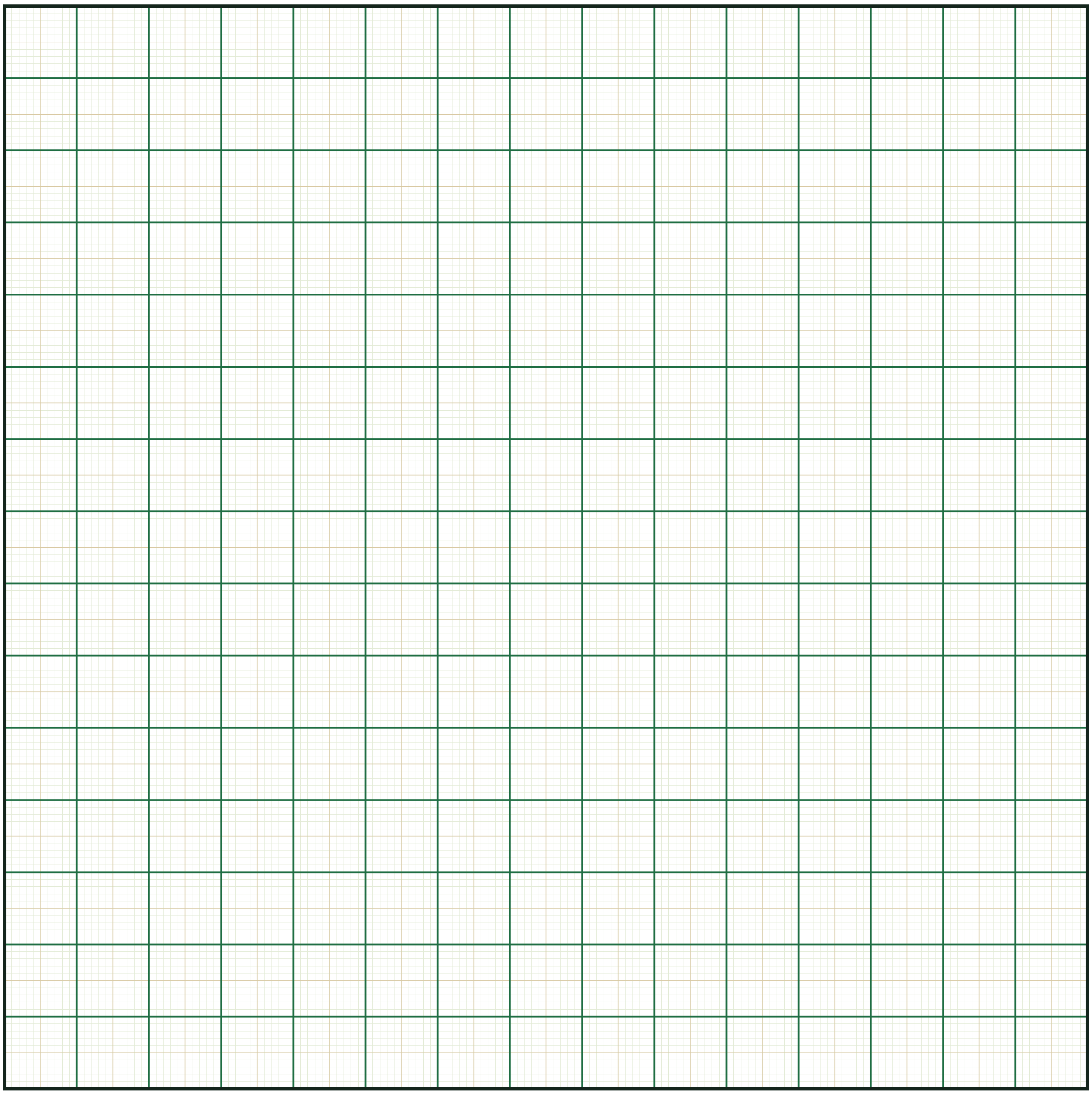
Observation Table

S. No.	b (kg/s)	m (g)	k_spring (N/m)	δ (log. decrement)	Q (quality factor)

S. No.	b (kg/s)	m (g)	k_spring (N/m)	δ (log. decrement)	Q (quality factor)

Graph

Logarithmic Decrement vs Damping Coefficient



Calculation

Result

b (kg/s)	δ	Q	T estimated (s)	T expected (s)

Maximum Permissible Error

Formula: $\Delta\delta = \Delta(\ln A_n) + \Delta(\ln A_{n+1}) \approx \Delta A_n/A_n + \Delta A_{n+1}/A_{n+1}$

Since δ is a logarithm of an amplitude ratio, its absolute error is approximately the sum of the fractional errors in the two amplitude readings — reading amplitudes several cycles apart and dividing by the cycle count reduces this error.

Quantity	Measured Value	Least Count (LC)
A_n — n-th peak amplitude (cm)		
$A_{(n+1)}$ — (n+1)-th peak amplitude (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is the ratio of successive amplitudes constant throughout an exponentially damped oscillation, rather than the difference?

2. Derive, in outline, the approximate relation $Q \approx \pi/\delta$ for light damping.

3. Why might you prefer to measure amplitudes several cycles apart rather than just two successive peaks?

4. How would critically damped or overdamped motion differ qualitatively from the lightly damped case studied here?

5. Give an example of a real system where a high Q is desirable, and one where a low Q is deliberately engineered.

6. How does the period of a lightly damped oscillator compare with the period of the same system if it were undamped?

14. Forced Oscillations and Resonance — Barton's Pendulums

Aim

To study the steady-state response of a damped oscillator to a periodic driving force, and to determine the resonant frequency and quality factor from the resonance curve.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$A(\omega) = (F_0/m) / \sqrt{[(\omega_0^2-\omega^2)^2+(2\gamma\omega)^2]} \; ; \; Q=f_0/\Delta f$$

ω, ω_0 = driving frequency and natural frequency
γ = damping constant
Δf = full width of the resonance curve at half-power

Labelled diagram of the experimental apparatus:

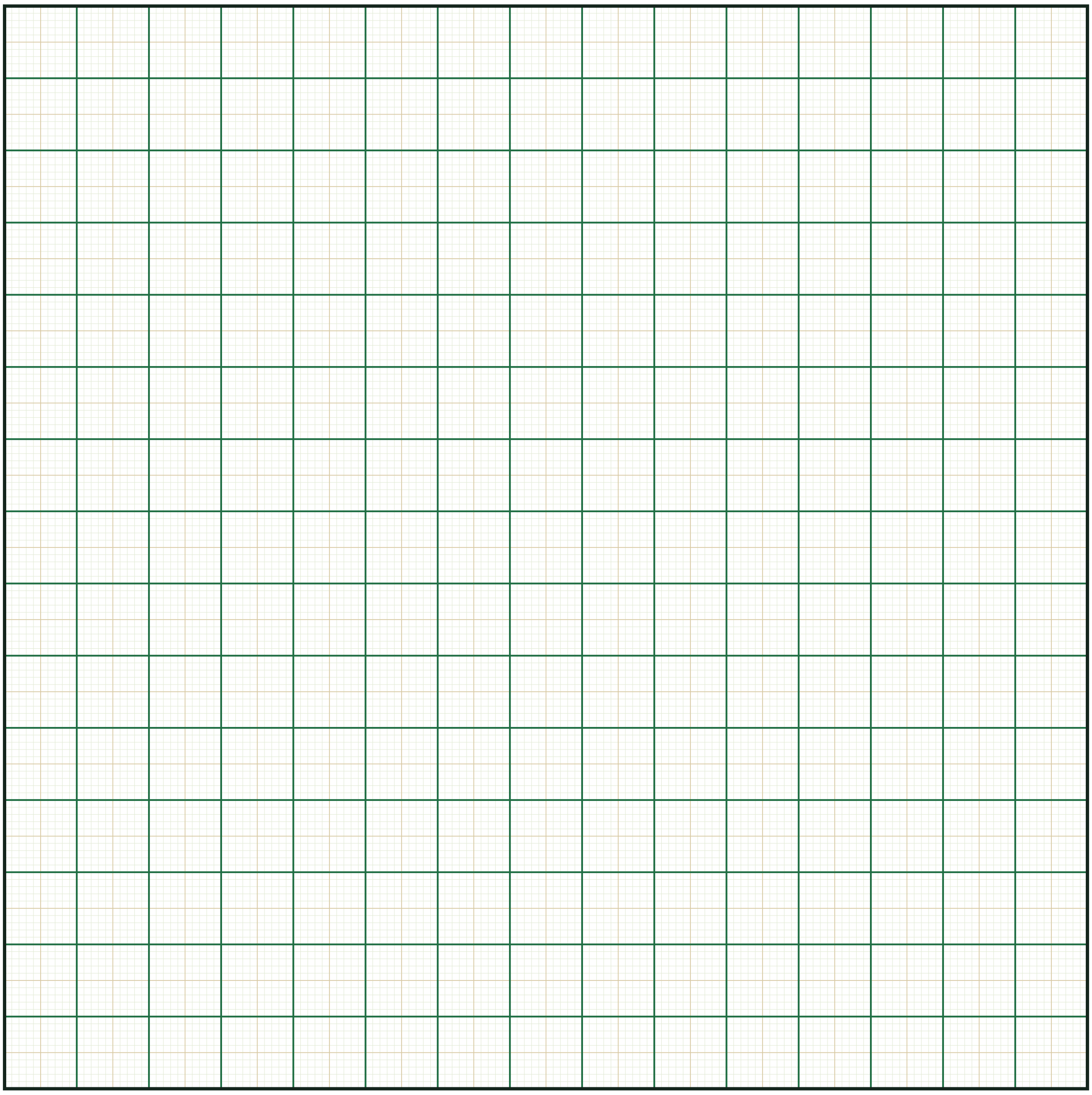
Observation Table

S. No.	f_drive (Hz)	Amplitude A (cm)	Phase lag φ (°)

S. No.	f_drive (Hz)	Amplitude A (cm)	Phase lag φ (°)

Graph

Resonance Curve



Amplitude, A (cm)

Driving frequency, f_drive (Hz)

Calculation

Result

γ (s^{-1})	f0 (Hz), from peak	Δf (Hz)	Q = f0/Δf

Maximum Permissible Error

Formula: $\Delta Q/Q = \Delta f_0/f_0 + \Delta(\Delta f)/\Delta f$

The half-power width Δf is usually the less precisely determined quantity, since it depends on locating two points on a curve that may be sparsely sampled near the half-power level.

Quantity	Measured Value	Least Count (LC)
f0 — Resonant (peak) frequency (Hz)		
Δf — Half-power width (Hz)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does the phase lag between the driven oscillator and the driving force pass through 90° exactly at resonance?

2. How is the quality factor obtained from a resonance curve related to the quality factor obtained from a free (damped) decay measurement?

3. In Barton's pendulums, why does only the pendulum whose length matches the driver's length swing with a large amplitude?

4. Why must you wait for the transient response to die away before reading a 'steady-state' amplitude at each driving frequency?

5. Give a real engineering example where avoiding resonance is a critical design consideration.

6. How would the resonance curve change if the damping were reduced to nearly zero?

15. Ultrasonic Interferometer — Velocity of Sound and Elastic Constants of a Liquid (PG)

Aim

To determine the velocity of ultrasonic waves in a liquid using an ultrasonic interferometer, and hence to calculate the liquid's adiabatic compressibility and bulk modulus.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\lambda = 2\Delta d \; ; \; v = f\lambda \; ; \; \beta = 1/(\rho v^2) \; ; \; K = \rho v^2$$

Δd = mean spacing between successive current maxima
f = crystal driving frequency
ρ = density of the liquid
β, K = adiabatic compressibility and bulk modulus

Labelled diagram of the experimental apparatus:

Observation Table

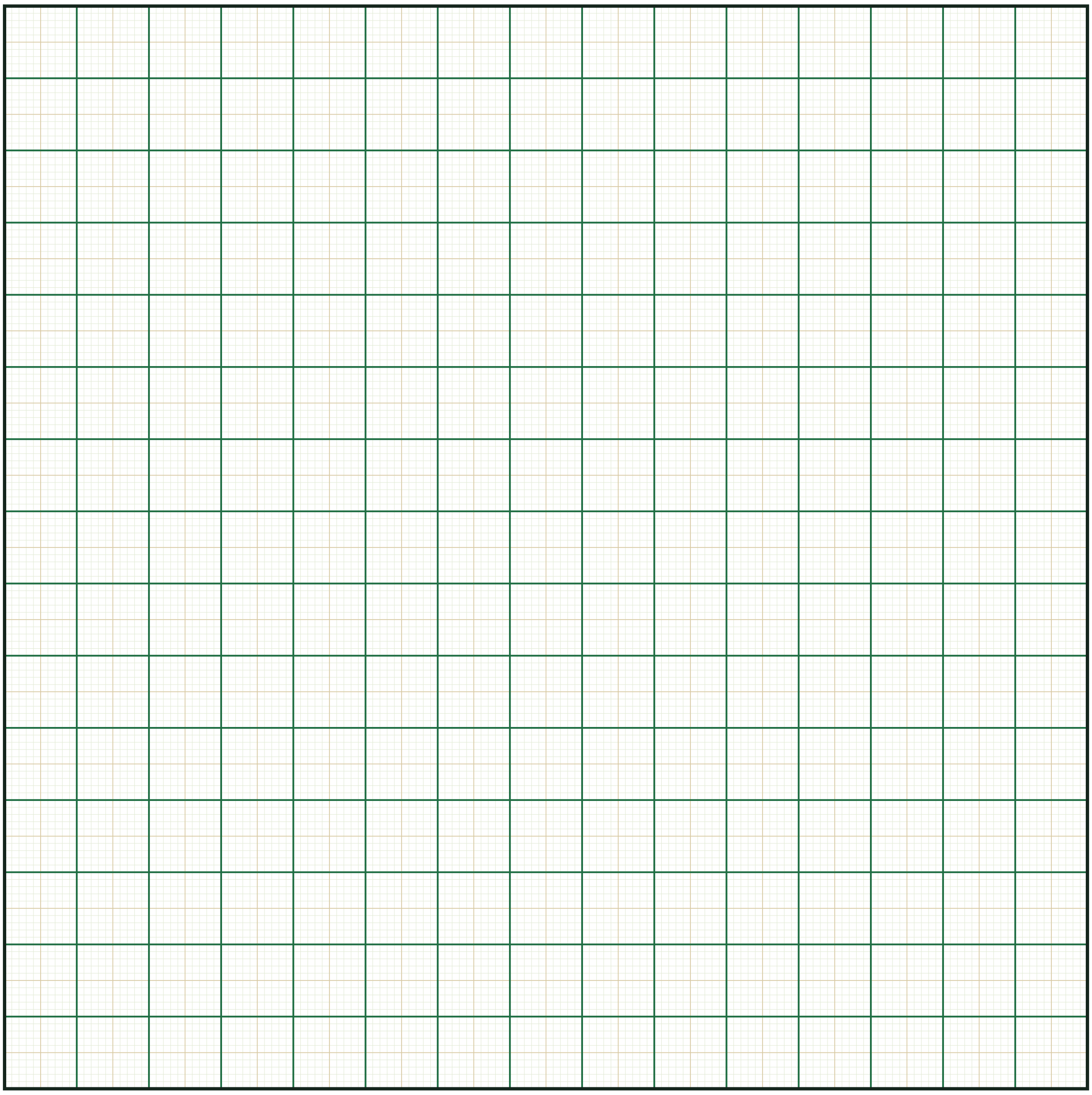
S. No.	Liquid	f (MHz)	N (maxima averaged)	Δd (mm)	$\lambda = 2\Delta d$ (mm)	$v = f\lambda$ (m/s)

S. No.	Liquid	f (MHz)	N (maxima averaged)	Δd (mm)	λ = 2Δd (mm)	v = fλ (m/s)

Graph

Total Distance vs Number of Maxima

Total distance moved, D (mm)



Number of maxima counted, N

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Calculation

Result

Liquid	ρ (kg/m^3)	v (m/s)	β (m^2/N)	K (N/m^2)

Maximum Permissible Error

Formula: $\Delta v/v = \Delta f/f + \Delta(\Delta d)/\Delta d$

Averaging the maxima spacing over as many resonances as practical (large N) directly reduces the fractional error in Δd, and hence in v, β and K, since micrometer positioning error is essentially fixed per reading but is divided by N when averaged this way.

Quantity	Measured Value	Least Count (LC)
f — Driving frequency (MHz)		
Δf — Frequency uncertainty (MHz)		
Δd — Mean maxima spacing (mm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Explain how a standing wave in the liquid column produces a sharp maximum in the crystal's drive current.

2. Why is it better to measure the total distance over many maxima (large N) and divide, rather than reading a single λ/2 spacing?

3. Derive the relations for adiabatic compressibility and bulk modulus from the measured velocity and known density.

4. Why does dissolved air or a temperature change affect the measured velocity of sound in a liquid?

5. What advantage does the ultrasonic interferometer have over other methods for finding the elastic constants of a liquid?

6. How would the observed velocity differ between a liquid and, say, air, and what does this reveal about the relative compressibility of liquids versus gases?

16. Ballistic Pendulum — Measuring an Unknown Launch Speed

Aim

To determine the launch speed of a projectile (ball) using a ballistic pendulum, by combining conservation of momentum during the (inelastic) impact with conservation of energy during the subsequent swing.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$v = [(M+m)/m]\cdot\sqrt{2gh}\ ;\ 1-\cos\theta = m^2v^2/[2gL(M+m)^2]$$

m, M = projectile (ball) mass, and pendulum block mass
h = vertical rise of the block+ball after impact
L = pendulum length
θ = maximum swing angle

Labelled diagram of the experimental apparatus:

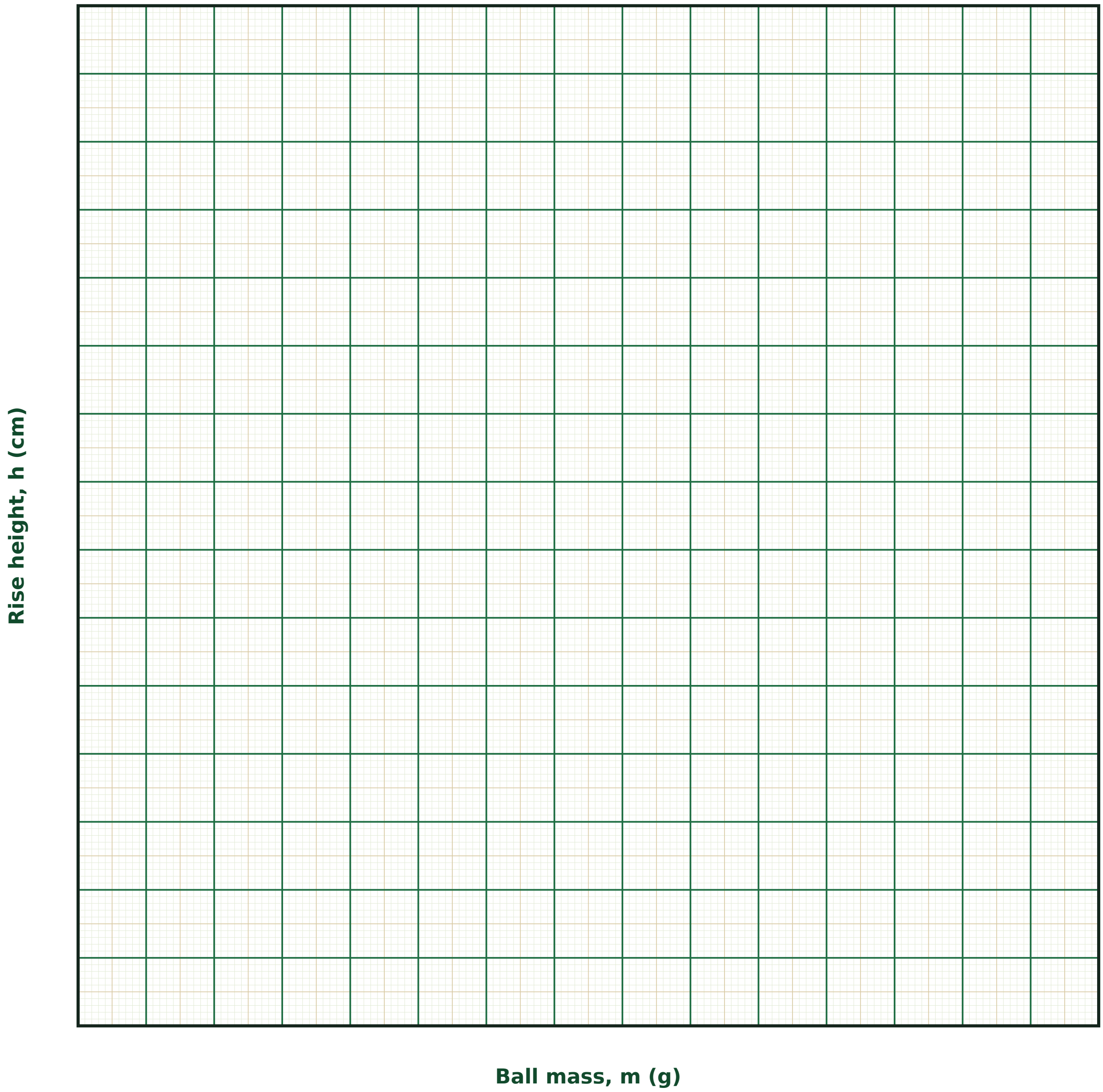
Observation Table

S. No.	m (g)	M (g)	L (cm)	θ_max (°)	h (cm)	Reconstructed v (m/s)

S. No.	m (g)	M (g)	L (cm)	θ_max (°)	h (cm)	Reconstructed v (m/s)

Graph

Rise Height vs Ball Mass



Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta v/v = \Delta m/m + \Delta M/M + \Delta L/L + \frac{1}{2}\cdot\Delta h/h$

$v = [(M+m)/m]\cdot\sqrt{2gh}$ depends on h only through a square root, so h contributes half its fractional error, while the mass ratio $(M+m)/m$ contributes its own fractional errors directly.

Quantity	Measured Value	Least Count (LC)
m — Ball (projectile) mass (g)		
M — Pendulum block mass (g)		
L — Pendulum length (cm)		
h — Rise height (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why must momentum conservation and energy conservation be applied in two separate stages here, rather than one energy-conservation equation for the whole process?

2. What physically happens to the "missing" kinetic energy that is not accounted for by energy conservation during the collision itself?

3. How would the result change if the ball only partially embedded and continued moving after passing through the block?

4. Why is it important to release the ball with a purely horizontal velocity, exactly at the pendulum's rest height?

5. Give one real-world application of the general technique (fast-collision-then-observe-the-slow-aftermath) used here.

6. How would friction at the pendulum's pivot bias your calculated launch speed — would it read too high or too low?

17. Rolling Race on an Incline — Moment of Inertia and Rolling Efficiency

Aim

To study how the mass distribution of a rolling body (characterized by its moment-of-inertia factor k) governs its acceleration down an incline, and to verify the theoretical prediction that a solid disc or sphere always outpaces a ring or hoop released from rest at the same height.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$a = g\sin\theta/(1+k)\ ;\ I = k\,m\,r^2\ ;\ t = \sqrt{2L/a}$$

θ = incline angle
k = moment-of-inertia factor (1 for a ring, ½ for a disc, ⅖ for a solid sphere, ⅔ for a hollow sphere)
L = distance travelled along the slope
t = time taken to reach the bottom

Labelled diagram of the experimental apparatus:

Observation Table

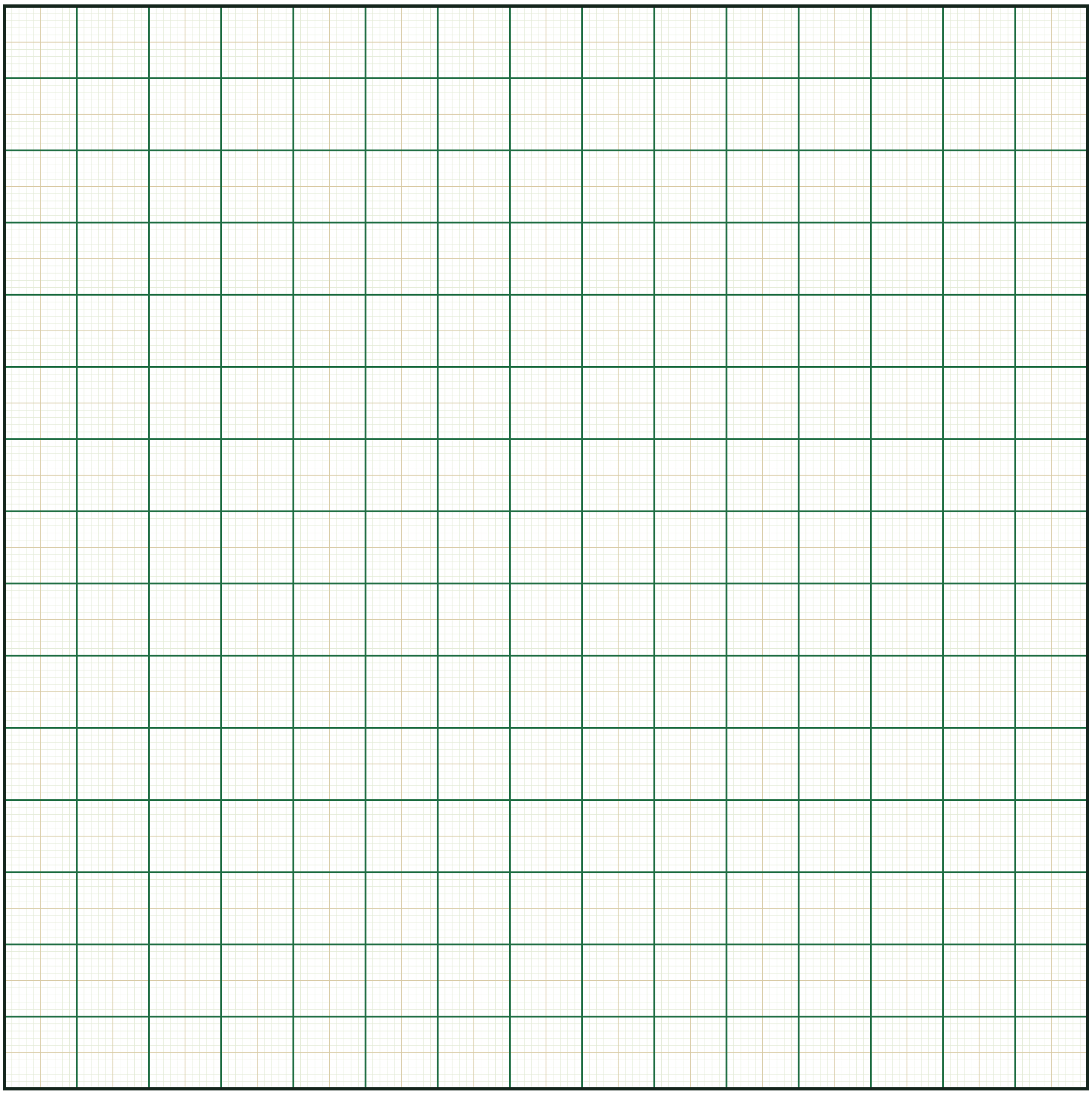
S. No.	Shape (k)	θ (°)	L (cm)	t (s)	a = $2L/t^2$ (cm/s²)	k (experimental)

S. No.	Shape (k)	θ (°)	L (cm)	t (s)	$a = 2L/t^2$ (cm/s^2)	k (experimen tal)

Graph

Time to Bottom vs Incline Angle

Time to reach bottom, t (s)



Incline angle, θ (°)

Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta k/(1+k) = \Delta a/a + \Delta\theta/\theta$ (small-angle approx.) ; a from $2\Delta s/t^2$ gives $\Delta a/a = \Delta L/L + 2\cdot\Delta t/t$

The moment-of-inertia factor k is inferred from the measured acceleration $a = g \sin\theta/(1+k)$; since a is obtained from the timed run ($a = 2L/t^2$), the fall time t enters squared and usually dominates the error budget.

Quantity	Measured Value	Least Count (LC)
L — Incline length (along slope) (cm)		
θ — Incline angle (°)		
t — Time to reach bottom (s)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is "rolling without slipping" essential to the formula used here, and what would go wrong if the body slid instead?

2. Explain, in terms of energy sharing, why a hoop always loses a rolling race to a solid disc of the same mass and radius.

3. How could you experimentally verify that a given ball is solid rather than hollow, using only this incline and a stopwatch?

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4. Why is the mass of the rolling body absent from the final formula for acceleration, even though gravity clearly depends on mass?

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5. Name one piece of real machinery (a flywheel, a vehicle wheel, an industrial roller) where the designer's choice of k has a genuine practical consequence, and explain why.

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6. How would rolling friction (a small resistive torque at the contact point) bias your experimentally determined k — too high or too low?

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18. Gyroscopic Precession — Spin, Torque and Steady Precession

Aim

To study the steady precession of a rapidly spinning wheel under an applied gravitational torque, and to verify that the precession rate is inversely proportional to the spin rate.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$L = I\omega \; ; \; \tau = Mgd \; ; \; \Omega = \tau/L = 2gd/(R^2\omega)$$

M, R = spinning wheel's mass and radius
ω = spin rate of the wheel about its own axle
d = offset of the wheel's centre of gravity from the pivot
Ω = steady precession rate of the axle about the vertical

Labelled diagram of the experimental apparatus:

Observation Table

S. No.	M (g)	R (cm)	d (cm)	ω (rad/s)	Ω (rad/s)	Ω·ω (check, ≈ constant)

S. No.	M (g)	R (cm)	d (cm)	ω (rad/s)	Ω (rad/s)	Ω·ω (check, ≈ constant)

Graph

Precession Rate vs Spin Rate

Precession rate, Ω (rad/s)

Spin rate, ω (rad/s)

Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta\Omega/\Omega = \Delta M/M + \Delta d/d + 2\cdot\Delta R/R + \Delta\omega/\omega$ (M cancels in the true formula, but propagate it if measured)

Since $\Omega = 2gd/(R^2\omega)$, the wheel radius R enters squared — the least-count error of whatever instrument measures R (often a vernier caliper) usually dominates the overall uncertainty in Ω.

Quantity	Measured Value	Least Count (LC)
M — Wheel mass (g)		
R — Wheel radius (cm)		
ω — Spin rate (rad/s)		
d — C.G. offset from pivot (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does an applied torque on a spinning gyroscope cause it to precess sideways rather than simply falling over in the direction of the torque?

2. Derive, in outline, why the precession rate Ω is inversely proportional to the spin rate ω.

3. Why does the wheel's mass drop out of the precession formula for a disc, and would it drop out for a different mass distribution (e.g., a hoop)?

4. Give one real engineering application where gyroscopic precession is deliberately exploited (not merely tolerated), and explain the underlying principle.

5. How would you expect the results to change if the wheel were spun in the opposite direction?

6. What would happen to the precession rate if the spin rate ω were allowed to slowly decay due to friction, over a long observation time?